

Predicting Pasture Biomass from Images: A Competition Report on the CSIRO Image2Biomass Kaggle Challenge

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Abstract

This report summarizes my participation in the CSIRO Image2Biomass Kaggle challenge [1], where the goal was to estimate pasture biomass from top-view images. The task is difficult because only 357 labeled training images are available, the five targets are partially constrained by physical relationships, and the public and private leaderboard splits exhibit noticeable distribution shift. I built an image-centric pipeline based on frozen visual foundation models (ConvNeXt-XLarge and DINOv2 Large/Giant), followed by PCA and support vector regression with Bayesian hyperparameter optimization. Across submitted variants, my best public leaderboard score reached 0.70 weighted R^2 , while the strongest private score reached 0.58. The main lessons were that DINOv2 features outperform a strong ConvNeXt baseline, patch-wise processing helps capture local vegetation heterogeneity, and blocked validation by sampling date is essential to avoid optimistic estimates.

1. Challenge and data

The competition asks participants to predict five biomass quantities for each pasture image: `Dry_Green_g`, `Dry_Dead_g`, `Dry_Clover_g`, `GDM_g`, and `Dry_Total_g`. The training set contains 357 top-view images, while the hidden test set contains more than 800 images. Each train image is accompanied by metadata such as sampling date, Australian state, species composition, NDVI, and mean vegetation height. The metadata are not available for test images so in this project I focused primarily on image-derived representations and used the sampling date mainly to build a safer validation scheme.

Performance is evaluated with a global weighted R^2 over all image-target pairs. The weights are 0.1, 0.1, 0.1, 0.2, and 0.5 for green, dead, clover, GDM, and total biomass respectively. Two targets are deterministic combinations of the others: $\text{GDM} = \text{Dry_Green} + \text{Dry_Clover}$ and $\text{Dry_Total} =$

Pipeline overview. Native-resolution image $\rightarrow$ frozen ConvNeXt/DINOv2 backbone $\rightarrow$ global or patch-wise features $\rightarrow$ PCA (50 components) $\rightarrow$ three RBF-SVRs for green, dead, and clover biomass $\rightarrow$ non-negativity clipping and reconstruction of GDM and total biomass.

Figure 1: The final system combined foundation-model representations with classical regression.

Dry_Green + Dry_Dead + Dry_Clover. This structure matters both for modeling and for evaluation, because prediction errors on the base components propagate to the derived targets.

The main technical challenges were the small number of labeled images, the high image resolution (approximately 1000×2000 pixels), and the strong spatial heterogeneity within each photograph: vegetation density, clover coverage, bare soil, and dry material can all vary substantially across the same frame.

2. Method

Because end-to-end deep learning would likely overfit on only 357 training images, I adopted a transfer-learning strategy in which large pretrained vision models were used only as *frozen feature extractors*. I tested two families of backbones. The first was ConvNeXt-XLarge [2], from which I extracted a global average pooled feature vector. The second was DINOv2 [3], using both the Large and Giant variants. For DINOv2 I concatenated two complementary representations: the global [CLS] embedding and the average of the patch-token embeddings.

To better exploit the panoramic structure of the pasture images, I extended the DINOv2 pipeline with patch-wise processing. Instead of encoding the whole image once, I split each image into regular grids of 2, 4, 8, or 16 tiles and processed every tile independently. I then aggregated the tile outputs in two ways: (i) by concatenating the average of the tile-level [CLS] vectors and the average of all patch tokens, and (ii) by collecting all patch tokens into a global “bag of

tokens” and computing summary statistics such as mean, standard deviation, skewness, kurtosis, GeM pooling, min/max, quantiles, positive-activation ratio, and L1/L2 norms. This produced richer descriptors for local texture and composition.

The extracted features were high-dimensional, so I removed constant columns and projected them to the first 50 principal components. On top of these compact representations, I trained three separate support vector regressors [4] with RBF kernels for `Dry_Green_g`, `Dry_Dead_g`, and `Dry_Clover_g`. The derived targets `GDM_g` and `Dry_Total_g` were reconstructed analytically from those three predictions, which reduced the number of models and guaranteed physically consistent outputs.

Hyperparameters (C, γ, ϵ) were tuned with Bayesian optimization [5]. To reduce data leakage, I used 4-fold blocked cross-validation where all samples collected on the same `Sampling_Date` were assigned to the same fold. This choice is important because images acquired on the same date often come from very similar field conditions; splitting them across train and validation would lead to overly optimistic estimates. I also clipped negative predictions to zero, which is a simple but useful constraint for biomass regression.

3. Results and discussion

The final submission placed 1113/3804 teams on the private leaderboard. Table 1 summarizes representative leaderboard results. The clearest pattern is that DINOv2 consistently outperformed ConvNeXt. Using DINOv2-Large on the whole image improved the public leaderboard from 0.58 to 0.65 relative to ConvNeXt-XLarge. Moving from whole-image encoding to patch-wise encoding further improved the public score up to 0.69, confirming that local structure matters for this problem.

Model	Public	Private
ConvNeXt-XL (full image)	0.58	0.54
DINOv2-L (full image)	0.65	0.53
DINOv2-L (2 tiles)	0.66	0.58
DINOv2-L (8 tiles)	0.69	0.56
DINOv2-G (8 tiles)	0.70	0.54
DINOv2-G (8 tiles + token stats)	0.70	0.57
Competition best	0.81	0.67

Table 1: Weighted R^2 on representative leaderboard submissions. The best public and private scores obtained across my submitted variants are highlighted.

story. The strongest private result, 0.58, came from a DINOv2-Large model with only two spatial tiles rather than from the most elaborate feature-engineering pipeline. Conversely, the DINOv2-Giant models achieved the best public score (0.70) but slightly lower private performance (0.54–0.57). This gap suggests a meaningful distribution shift between the public and private splits and shows that aggressive tuning against the public leaderboard can be misleading.

The project also exposed several practical implementation issues. First, extracting features at native resolution from large backbones is computationally expensive; using Kaggle’s Tesla P100 GPUs was essential. Second, a naive random validation split would have been unreliable because same-date samples are too similar. Third, richer local descriptors are not automatically better: adding more tiles and more summary statistics increases representation capacity, but it can also amplify variance when the labeled set is very small.

Overall, the system behaved like a strong small-data baseline: foundation-model features captured useful visual cues without finetuning, while SVR provided a robust regression layer on top of compact PCA embeddings.

4. Conclusion

This competition taught me how to turn modern visual foundation models into practical predictors for a scientific regression task with limited supervision. The most valuable ideas were to preserve image resolution, model only the independent targets, and design validation splits that reflect the data-collection process rather than random sampling. In future work I would combine image features with the available meta-data (for example NDVI and pasture height), explore lightweight finetuning with data augmentation, and build ensembles selected by blocked cross-validation.

The private leaderboard tells a more nuanced

References

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- [4] Alex J. Smola and Bernhard Schölkopf. A Tutorial on Support Vector Regression. *Statistics and Computing*, 14(3):199–222, 2004.
- [5] Bernd Bischl, Jakob Richter, Janek Thomas, Michel Lang, and Florian Hutter. mlrMBO: A Modular Framework for Model-Based Optimization of Expensive Black-Box Functions. *arXiv preprint arXiv:1703.03373*, 2017.

A. Additional implementation details

This appendix gives extra information on the submitted pipeline.

A.1. Feature extraction at native resolution

To preserve fine-grained cues such as grass texture, dry material, clover patches, and exposed soil, I processed the images at their original resolution of approximately 1000×2000 pixels. ConvNeXt-XLarge was used as a frozen ImageNet backbone with native Keras preprocessing and global average pooling, yielding a 1536-dimensional descriptor. DINOv2 ViT-L/14 and ViT-g/14 were loaded in PyTorch from local Hugging Face checkpoints. For whole-image DINOv2 encoding, I concatenated the global [CLS] embedding with the mean of all patch tokens, giving a descriptor of length 2048 for DINOv2-L and 3072 for DINOv2-G. Running these models at native resolution required Kaggle’s NVIDIA Tesla P100 GPUs.

A.2. Patch-wise aggregation

Because DINOv2 is a Vision Transformer, each image (or tile) is represented by two complementary outputs: a single [CLS] token and a set of patch-level tokens, one for each 14×14 pixels patch of the tile. The [CLS] token acts as a global summary of the visual content, aggregating information from the entire image into one embedding, whereas the patch tokens retain spatially distributed information about local regions. In practice, this means that the [CLS] embedding is more suitable for capturing overall image context, while the patch tokens provide a richer description of fine-scale texture and structure. To leverage both sources of information, I explored two tiled DINOv2 strategies. The simpler one split each image into regular grids of 2, 4, 8, or 16 tiles and concatenated the averaged tile-level [CLS] embeddings and the mean of all patch tokens. The richer variant additionally pooled all patch tokens from all tiles into a global “bag of tokens” and computed channel-wise statistics: mean, standard deviation, skewness, kurtosis, GeM pooling with $p = 3$, minimum, maximum, quantiles at 0.10/0.25/0.50/0.75/0.90, the proportion of positive activations, the L1 norm, the L2 norm, and the L1/L2 ratio. Before regression, I removed zero-variance columns and retained the first 50 principal components after PCA with scaling.

A.3. Regression setup and SVR Hyperparameters optimization

On top of the 50-dimensional PCA representation, I trained three separate support vector regressors

with radial basis function (RBF) kernels, one for each independent biomass component: `Dry_Green_g`, `Dry_Dead_g`, and `Dry_Clover_g`. I chose to model these targets separately because they represent the fundamental components of the biomass decomposition, whereas `GDM_g` and `Dry_Total_g` are deterministic sums of them. This design reduced the number of supervised models to be estimated, allowed each regressor to adapt its own hyperparameters to the statistical properties of its target, and ensured that the final predictions remained physically consistent once the derived targets were reconstructed analytically. Hyperparameters were tuned independently for each target using Bayesian optimization, treating the cross-validated SVR performance as a black-box objective function. For each candidate triplet (C, γ, ϵ) , an RBF-SVR was trained and evaluated with 4-fold blocked cross-validation, where all samples sharing the same `Sampling_Date` were kept in the same fold to reduce temporal leakage. After each evaluation, the optimizer updated a surrogate model of the relationship between hyperparameters and validation performance, and used this model to propose the next configuration by balancing exploration of uncertain regions of the search space with exploitation of promising areas. The search was carried out over $\log_{10} C \in [-2, 4]$, $\log_{10} \gamma \in [-5, 1]$, and $\log_{10} \epsilon \in [-3, 0]$ for 80 iterations. This strategy is more sample-efficient than exhaustive grid search because each objective evaluation requires fitting and validating a full SVR model, while the response surface is highly non-linear and may contain narrow high-performing regions. The best hyperparameter setting for each target was then selected from the optimization trajectory and used to train the final model.

B. Extended results and observations

Variant	Public	Private
ConvNeXt-XL (whole image)	0.58	0.54
DINOv2-L (whole image)	0.65	0.53
DINOv2-L (2 tiles)	0.66	0.58
DINOv2-L (4 tiles)	0.68	0.57
DINOv2-L (8 tiles)	0.69	0.56
DINOv2-L (16 tiles)	0.67	0.55
DINOv2-G (8 tiles)	0.70	0.54
DINOv2-G (8 tiles + token stats)	0.70	0.57

Table A1: More complete leaderboard summary from the project archive. The simpler DINOv2-L model with two tiles produced the strongest private score, while the most elaborate DINOv2-G variants were best on the public split.

The appendix table makes two patterns clearer than the condensed main-paper table. First, tiling helped up to a point: moving from whole-image encoding to 4–8 tiles improved the public score, but pushing the decomposition to 16 tiles slightly reduced both public and private performance. This suggests a trade-off between capturing local heterogeneity and preserving enough global context. Second, the private leaderboard was maximized by a relatively simple DINOv2-L configuration rather than by the highest-capacity model, which reinforces the conclusion that the public leaderboard alone was a weak model-selection signal.

C. Interactive DINOv2-L features UMAP

As a final qualitative sanity check, the full project also included an interactive UMAP projection of whole-image DINOv2-L features. Visually similar pastures tended to cluster together, indicating that the frozen embeddings captured meaningful differences in vegetation composition and ground cover even before supervised regression. The interactive UMAP visualization HTML can be found at [this link](#).